

## T 5870 EN

### Series 240

**Type 3241/3374 Electric Control Valve; Type 3241 Globe Valve**

**Type 3244/3374 Electric Control Valve; Type 3244 Three-way Valve**



### Application

Versatile control valves for industrial applications as well as for heating, ventilation and air-conditioning systems

**DN 15 to 150 · PN 16 to 40 · Temperatures from -196 to +450 °C**

### Special features

Type 3244 Three-way Valve or Type 3241 Globe Valve with Type 3374 Electric Actuator

- Type 3241 Globe Valve
- Type 3244 Three-way Valve

### Versions

The Type 3374 Electric Actuator is available in various versions (for details, refer to associated data sheet):

- With additional electrical equipment (limit contacts, resistance transmitter, positioner)

### Options

- Electric control valves with other electric actuators, see associated data sheets
- Valves with pneumatic actuators
- Version tested according to DIN EN 14597 available

### Conformity



### Installation

The control valves can be installed in any desired position except with the actuator suspended downwards. The actuator is attached to the yoke with a fastening nut.



**Fig. 1:** Type 3241/3374



**Fig. 2:** Type 3244/3374

**Table 1:** Nominal sizes,  $K_{VS}$  coefficients, seat diameters and permissible differential pressures  $\Delta p$  in bar when  $p_2 = 0$  bar**Table 1.1:** Types 3241/3374-15, -17

| K <sub>VS</sub>                                    | 0.1 | 0.16 | 0.25 | 0.4 | 0.63 | 1.0 | 1.6 | 2.5 | 4.0 | 6.3 | 10 | 16   | 25   | 40   | 60  | 63  | 80  | 100  | 160  | 200 | 260 |     |
|----------------------------------------------------|-----|------|------|-----|------|-----|-----|-----|-----|-----|----|------|------|------|-----|-----|-----|------|------|-----|-----|-----|
| Seat Ø    mm                                       | 3   |      |      | 6   |      |     | 12  |     |     | 24  |    |      | 31   | 38   | 48  | 63  |     | 80   |      | 100 | 110 | 130 |
| Travel    mm                                       | 15  |      |      |     |      |     |     |     |     |     |    |      |      |      |     | 30  | 15  | 30   |      |     |     |     |
| DN                                                 |     |      |      |     |      |     |     |     |     |     |    |      |      |      |     |     |     |      |      |     |     |     |
| 15                                                 | •   | •    | •    | •   | •    | •   | •   | •   | •   |     |    |      |      |      |     |     |     |      |      |     |     |     |
| 20                                                 | •   | •    | •    | •   | •    | •   | •   | •   | •   | •   |    |      |      |      |     |     |     |      |      |     |     |     |
| 25                                                 | •   | •    | •    | •   | •    | •   | •   | •   | •   | •   | •  |      |      |      |     |     |     |      |      |     |     |     |
| 32                                                 |     |      |      | •   | •    | •   | •   | •   | •   | •   | •  | •    |      |      |     |     |     |      |      |     |     |     |
| 40                                                 |     |      |      | •   | •    | •   | •   | •   | •   | •   | •  | •    | •    |      |     |     |     |      |      |     |     |     |
| 50                                                 |     |      |      | •   | •    | •   | •   | •   | •   | •   | •  | •    | •    | •    |     |     |     |      |      |     |     |     |
| 65                                                 |     |      |      |     |      |     |     |     |     |     | •  | •    | •    | •    | •   |     |     |      |      |     |     |     |
| 80                                                 |     |      |      |     |      |     |     |     |     |     | •  | •    | •    | •    | •   |     | •   |      |      |     |     |     |
| 100                                                |     |      |      |     |      |     |     |     |     |     |    |      |      | •    |     | •   |     | •    | •    |     |     |     |
| 125                                                |     |      |      |     |      |     |     |     |     |     |    |      |      | •    |     | •   |     | •    | •    | •   |     |     |
| 150                                                |     |      |      |     |      |     |     |     |     |     |    |      |      | •    |     | •   |     | •    | •    |     | •   |     |
| Δp when p <sub>2</sub> = 0 bar                     |     |      |      |     |      |     |     |     |     |     |    |      |      |      |     |     |     |      |      |     |     |     |
| Without balanced plug                              |     |      |      |     |      |     |     |     |     |     |    |      |      |      |     |     |     |      |      |     |     |     |
| 3374-15                                            | 40  | 40   | 40   | 40  | 40   | 40  | 40  | 40  | 40  | 40  | 40 | 26.5 | 17.5 | 10.6 | 5.8 | 5.8 | 3.4 | 3.4  | 2.0  | 1.5 | 1.0 |     |
| 3374-17                                            | –   | –    | –    | –   | –    | –   | –   | –   | –   | –   | –  | 40   | 37   | 23   | 13  | 13  | 7.5 | 7.5  | 4.5  | 3.5 | 2.5 |     |
| With PTFE pressure balancing, without bellows seal |     |      |      |     |      |     |     |     |     |     |    |      |      |      |     |     |     |      |      |     |     |     |
| 3374-15                                            | –   | –    | –    | –   | –    | –   | –   | –   | –   | –   | –  | –    | –    | –    | –   | –   | 40  | 40   | 40   | 40  | 40  |     |
| 3374-17                                            | –   | –    | –    | –   | –    | –   | –   | –   | –   | –   | –  | –    | –    | –    | –   | –   | 40  | 40   | 40   | 40  | 40  |     |
| With PTFE pressure balancing, with bellows seal    |     |      |      |     |      |     |     |     |     |     |    |      |      |      |     |     |     |      |      |     |     |     |
| 3374-15                                            | –   | –    | –    | –   | –    | –   | –   | –   | –   | –   | –  | –    | –    | –    | –   | –   | 40  | 12.1 | 10.3 | 9.4 | 7.4 |     |
| 3374-17                                            | –   | –    | –    | –   | –    | –   | –   | –   | –   | –   | –  | –    | –    | –    | –   | –   | 40  | 34   | 32   | 30  | 28  |     |

**Table 1.2:** Types 3244/3374-15, -17

| K <sub>VS</sub>                | 2  | 4  | 6.3 | 10 | 6.3  | 10   | 16   | 25   | 25   | 40   | 60  | 80  | 100 | 140 | 160 | 200 | 300 |     |
|--------------------------------|----|----|-----|----|------|------|------|------|------|------|-----|-----|-----|-----|-----|-----|-----|-----|
| Seat Ø    mm                   | 24 |    |     |    | 31   |      |      | 38   | 48   |      |     | 63  | 75  | 80  | 90  | 100 | 110 | 130 |
| Travel    mm                   | 15 |    |     |    |      |      |      |      |      |      |     |     | 30  |     |     |     |     |     |
| DN                             |    |    |     |    |      |      |      |      |      |      |     |     |     |     |     |     |     |     |
| 15                             | •  | •  |     |    |      |      |      |      |      |      |     |     |     |     |     |     |     |     |
| 20                             | •  | •  | •   |    |      |      |      |      |      |      |     |     |     |     |     |     |     |     |
| 25                             | •  | •  | •   | •  |      |      |      |      |      |      |     |     |     |     |     |     |     |     |
| 32                             |    |    | •   | •  | •    | •    | •    |      |      |      |     |     |     |     |     |     |     |     |
| 40                             |    |    | •   | •  | •    | •    | •    | •    |      |      |     |     |     |     |     |     |     |     |
| 50                             |    |    | •   | •  | •    | •    | •    | •    |      | •    |     |     |     |     |     |     |     |     |
| 65                             |    |    |     |    |      |      |      |      | •    | •    | •   |     |     |     |     |     |     |     |
| 80                             |    |    |     |    |      |      |      | •    | •    | •    | •   | •   |     |     |     |     |     |     |
| 100                            |    |    |     |    |      |      |      |      |      |      |     |     | •   |     | •   |     |     |     |
| 125                            |    |    |     |    |      |      |      |      |      |      |     |     |     | •   |     | •   |     |     |
| 150                            |    |    |     |    |      |      |      |      |      |      |     |     |     |     |     | •   | •   |     |
| Δp when p <sub>2</sub> = 0 bar |    |    |     |    |      |      |      |      |      |      |     |     |     |     |     |     |     |     |
| 3374-15                        | 40 | 40 | 40  | 40 | 26.8 | 26.8 | 26.8 | 17.5 | 10.6 | 10.6 | 5.8 | 3.9 | 3.3 | 2.5 | 1.9 | 1.5 | 1.0 |     |
| 3374-17                        | 40 | 40 | 40  | 40 | 40   | 40   | 40   | 37   | 23   | 23   | 13  | 9   | 7.8 | 6   | 4.8 | 3.9 | 2.7 |     |

**Table 2:** Dimensions and weights**Table 2.1:** Type 3241 Valve (without actuator)

| Nominal size          | DN           | 15     | 20  | 25  | 32  | 40  | 50  | 65  | 80  | 100 | 125 | 150 |
|-----------------------|--------------|--------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Length L              | mm           | 130    | 150 | 160 | 180 | 200 | 230 | 290 | 310 | 350 | 400 | 480 |
| Height H1             | mm           | H2 + H |     |     |     |     |     |     |     |     |     |     |
| Height H2             | mm           | 222    | 222 | 222 | 223 | 223 | 223 | 262 | 262 | 354 | 363 | 390 |
| Height H3             | mm           | 61     | 61  | 61  | 61  | 61  | 61  | 61  | 61  | 75  | 75  | 75  |
| Height H4, fail-close | mm           | 75     | 75  | 75  | 75  | 75  | 75  | 75  | 75  | 90  | 90  | 90  |
| Height H5             | mm           | 44     | 44  | 44  | 72  | 72  | 72  | 98  | 98  | 118 | 144 | 175 |
| Weight                | kg (approx.) | 6      | 7.5 | 8   | 12  | 14  | 18  | 29  | 34  | 52  | 81  | 108 |

**Table 2.2:** Type 3241 Valve with insulating section or bellows seal (without actuator)

| Nominal size |                        |              | DN  | 15   | 20  | 25  | 32  | 40  | 50  | 65  | 80  | 100 | 125 | 150 |
|--------------|------------------------|--------------|-----|------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Height<br>H9 | Short/with bellows     | mm           | 409 | 409  | 409 | 410 | 410 | 410 | 410 | 451 | 451 | 636 | 645 | 672 |
|              | Long/long with bellows | mm           | 713 | 713  | 713 | 714 | 714 | 714 | 714 | 755 | 755 | 877 | 886 | 913 |
| Weight       | Short/with bellows     | kg (approx.) | 9   | 10.5 | 11  | 18  | 20  | 24  | 24  | 37  | 42  | 70  | 106 | 138 |
|              | Long/long with bellows | kg (approx.) | 13  | 14.5 | 15  | 22  | 24  | 28  | 28  | 41  | 46  | 78  | 114 | 146 |

**Table 2.3:** Type 3241 Valve with heating jacket (not for valve body made of EN-GJL-250 or EN-GJS-400-18-LT)

| Nominal size | DN | 25  | 32...50 | 65...80 | 100 |
|--------------|----|-----|---------|---------|-----|
| a            | mm | 110 | 140     | 180     | 200 |
| b            | mm | 15  | 20      | 35      | 50  |
| c            | mm | 140 | 170     | 215     | 255 |

Table 2.4: Type 3374 Actuator

|           |              |                                                                                                   |
|-----------|--------------|---------------------------------------------------------------------------------------------------|
| Height H  | mm           | 220                                                                                               |
| Height H6 | mm           | 150 (min. clearance required to disassemble the actuator)/300 (cover screws are mounted from top) |
| Weight    | kg (approx.) | 3.3                                                                                               |

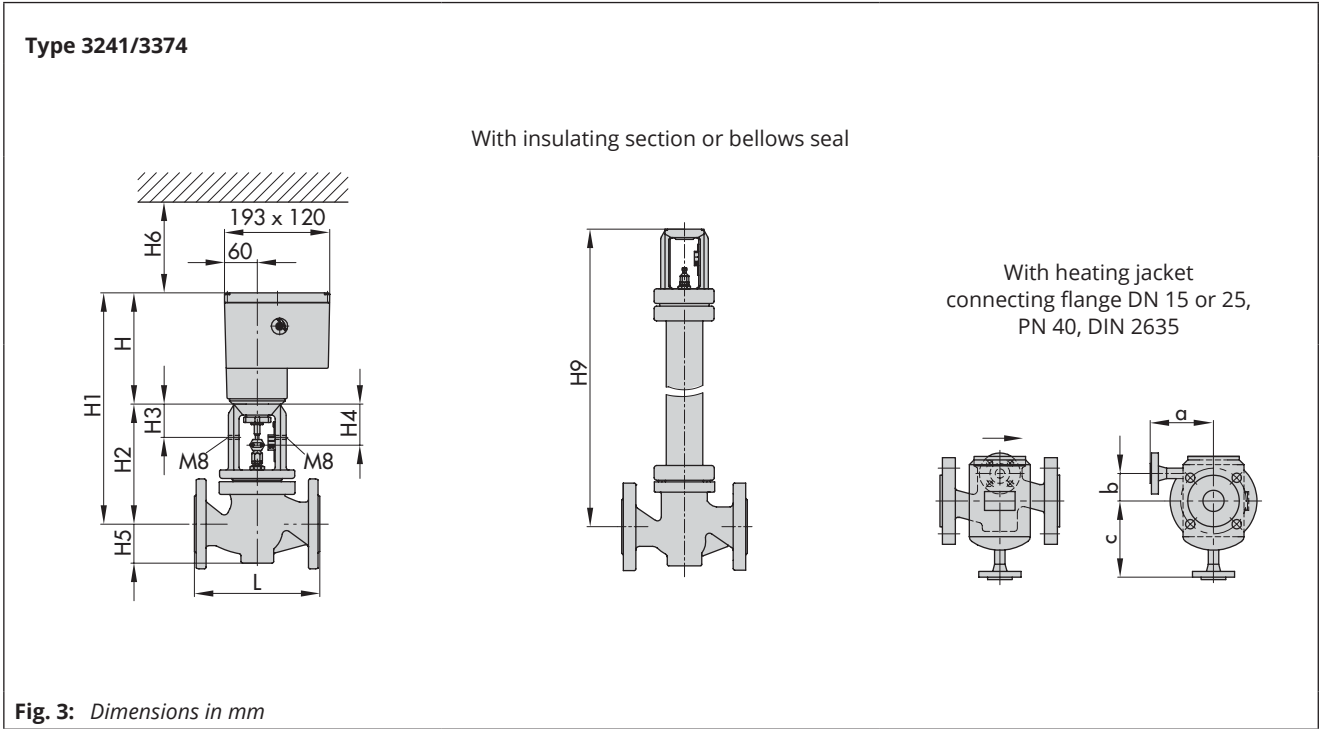


Table 2.5: Type 3244 Valve (without actuator)

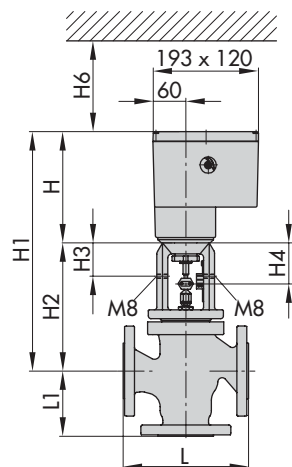
|                          |              |        |     |     |     |     |     |     |     |     |     |     |
|--------------------------|--------------|--------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Nominal size             | DN           | 15     | 20  | 25  | 32  | 40  | 50  | 65  | 80  | 100 | 125 | 150 |
| Length L                 | mm           | 130    | 150 | 160 | 180 | 200 | 230 | 290 | 310 | 350 | 400 | 480 |
| Height L1                | mm           | 70     | 80  | 85  | 100 | 105 | 120 | 130 | 140 | 150 | 200 | 210 |
| Height H1                | mm           | H2 + H |     |     |     |     |     |     |     |     |     |     |
| Height H2                | mm           | 235    | 235 | 235 | 235 | 235 | 235 | 270 | 270 | 360 | 375 | 375 |
| Height H3                | mm           | 61     | 61  | 61  | 61  | 61  | 61  | 61  | 61  | 75  | 75  | 75  |
| Height H4, stem extended | mm           | 75     | 75  | 75  | 75  | 75  | 75  | 75  | 75  | 90  | 90  | 90  |
| Weight                   | kg (approx.) | 6      | 7   | 8   | 13  | 15  | 17  | 31  | 37  | 49  | 95  | 135 |

Table 2.6: Type 3244 Valve with insulating section or bellows seal (without actuator)

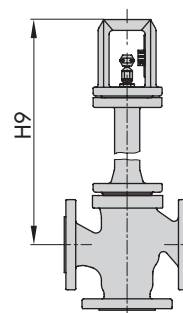
| Nominal size |                        | DN           | 15  | 20  | 25  | 32  | 40  | 50  | 65  | 80  | 100 | 125 | 150 |
|--------------|------------------------|--------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Height<br>H9 | Short/with bellows     | mm           | 420 | 420 | 420 | 420 | 420 | 420 | 455 | 455 | 645 | 655 | 655 |
|              | Long/long with bellows | mm           | 725 | 725 | 725 | 725 | 725 | 725 | 760 | 760 | 895 | 900 | 900 |
| Weight       | Short/with bellows     | kg (approx.) | 9   | 10  | 11  | 19  | 21  | 23  | 40  | 45  | 68  | 120 | 165 |
|              | Long/long with bellows | kg (approx.) | 13  | 14  | 15  | 23  | 25  | 27  | 44  | 49  | 76  | 128 | 173 |

**Table 2.7:** *Type 3374 Actuator*

|           |              |                                                                                                   |
|-----------|--------------|---------------------------------------------------------------------------------------------------|
| Height H  | mm           | 220                                                                                               |
| Height H6 | mm           | 150 (min. clearance required to disassemble the actuator)/300 (cover screws are mounted from top) |
| Weight    | kg (approx.) | 3.3                                                                                               |

**Type 3244/3374**

With insulating section or bellows seal

**Fig. 4:** *Dimensions in mm*

### Ordering text

- Electric control valve · Type 3241/3374 or Type 3244/3374
- DN ..., body material, PN ...
- For Type 3244 Valve: mixing or diverting valve
- Type 3374 Actuator:
  - Supply voltage of version with three-step signal:
    - 230 V, 50 Hz
    - 24 V, 50 Hz
  - Supply voltage of version with positioner:
    - 85 to 264 V/50 and 60 Hz
    - 24 V/50 and 60 Hz and DC
- Optionally, special version

### Associated data sheets

- ▶ T 8015
- ▶ T 8026
- ▶ T 8331

### Associated mounting and operating instructions

- ▶ EB 8015
- ▶ EB 8026
- ▶ EB 8331-3
- ▶ EB 8331-4